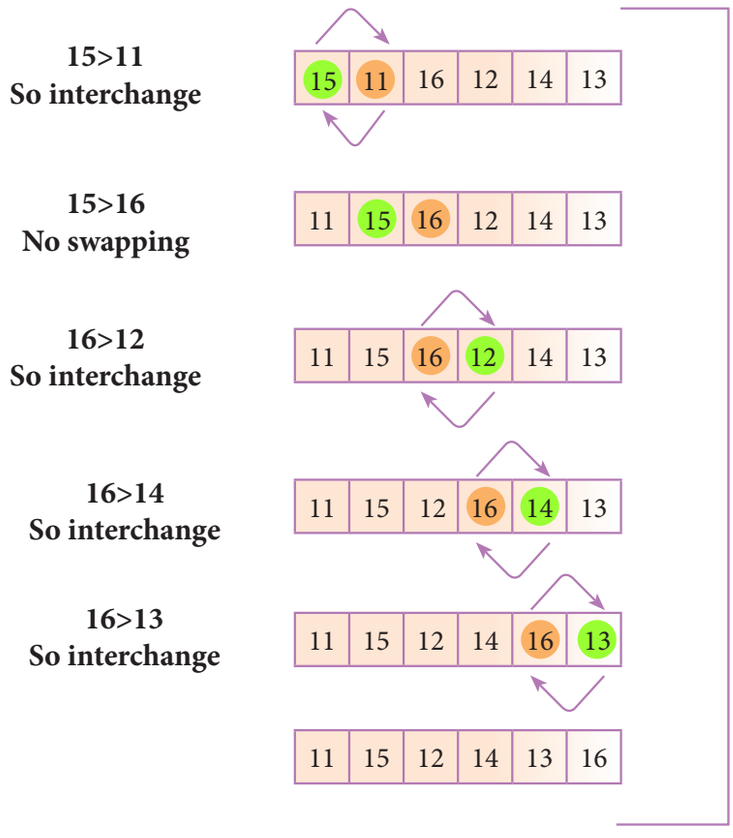
Let's consider an array with values {15, 11, 16, 12, 14, 13} Below, we have a pictorial representation of how bubble sort will sort the given array.



The above pictorial example is for iteration-1. Similarly, remaining iteration can be done. The final iteration will give the sorted array.

At the end of all the iterations we will get the sorted values in an array as given below:

11	12	13	14	15	16

4.5.2 Selection sort

The selection sort is a simple sorting algorithm that improves on the performance of bubble sort by making only one exchange for every pass through the list. This algorithm will first find the smallest elements in array and swap it with the element in the first position of an array, then it will find the second smallest element and swap that element with the element in the second position, and it will continue until the entire array is sorted in respective order.

This algorithm repeatedly selects the next-smallest element and swaps in into the right place for every pass. Hence it is called selection sort.

Procedure

1. Start from the first element i.e., index-0, we search the smallest element in the array, and replace it with the element in the first position.

12th Computer Science_EM Chapter 4.indd 39 21-12-2022 15:55:00